

Mohanad Ayoub

✉ mohanadayoubx@gmail.com ☎ +967 772 446 214 📍 Yemen, Taiz

🔗 mohanad-ayoub.netlify.app 🌐 github.com/MohanadX

🌐 linkedin.com/in/mohanad-ayoub-55bb29382

Summary

Next.js & React Frontend Developer specializing in high-performance, full-stack web applications. Expert in leveraging modern rendering strategies (SSR/ISR) to optimize SEO and Core Web Vitals. Proven track record of delivering seamless user experiences through optimistic UI updates, robust state management, and resilient error-handling architectures.

Skills

Languages & Core:

TypeScript, JavaScript (ES6+), Python

Backend:

Next.js API Routes, MongoDB with Mongoose, REST APIs, ESLint, Prettier, Zod, Trigger.dev [🔗](#), Prisma, Drizzle ORM, Arcjet and ImageKit, AI/LLM Integration: Gemini API

Frontend:

React 19, Next.js (SSR), Tailwind CSS, Shadcn/UI, Material UI, Redux Toolkit, React Query, Context API

Tools & Operations:

Jest, React testing library, Next-Auth (v5), Clerk, Arcjet, Sentry, PostHog, Git/GitHub

Projects

Flow AI [🔗](#)

- Developed Flow AI, a real-time collaborative system architecture platform
- **Reduced page load times by 60%** by architecting a client-side caching layer with React Query and implementing code-splitting/lazy-loading for complex visual canvas components.
- Implemented Websockets (Pusher) for real-time updates
- Engineered a real-time collaborative canvas using React Flow powered with LiveBlocks backend allowing concurrent users to design software architectures simultaneously.
- Handled long task run via Trigger.dev backend including AI spec-generation and AI design chat with comprehensive request live state feed for user
- Tech stack: React 19, Next.js, TypeScript, Tailwind CSS, React Flow, Prisma, Liveblocks, Pusher, Clerk, Trigger.dev [🔗](#), Vercel, Gemini API

ECourse Platform [🔗](#)

- Built an LMS platform managing courses and products for Admins and users
- Devised an advanced hybrid pagination strategy combining Next.js server-side caching for initial page loads with React Query client-side caching for subsequent data requests. This reduced server requests significantly while gracefully handling infinite loading entirely on the client, removing server-side memory limits.
- **Integrated Clerk** to manage secure, multi-role user authentication and session management for administrators and users.
- Architected a secure LMS database schema utilizing **PostgreSQL** and **Drizzle ORM**, optimizing query performance for course progress tracking.
- Implemented a dynamic, location-aware pricing system using **Arcjet**, automatically applying Purchasing Power Parity (PPP) discounts based on country-level IP data.
- Tech stack: React 19, Next.js 16 (App Router), Tailwind CSS, shadcn/ui, dnd kit, PostgreSQL, Drizzle ORM, Stripe, Arcjet, Clerk, ImageKit, Vercel, Docker

Lorial

- **Boosted data load efficiency by 50%** across a developer event platform by implementing an advanced hybrid caching strategy using React Query and Next.js server-side caching.
- Managed cloud image processing via ImageKit and MongoDB Atlas.
- Integrated **PostHog** to establish real-time observability, reducing error-identification time and capturing user behavior analytics.
- Implemented robust authentication using Next Auth, integrating OAuth and credential providers for secure user access.
- React 19, Next.js 16, React Query, MongoDB Atlas, Next-Auth, ImageKit, Zod, PostHog

Certificates

- **CS50P:**  CS50's Introduction to Programming with Python, Harvard University (via edX). Completed on 2024. Mastered Python programming, algorithms, and data structures through project-based learning.
- **Algorithms and Data Structures:**  Mastering Algorithms and Data Structures in TypeScript with Codesignal web platform with problem-based learning. Completed on 2025
- **Micro1 Certification:**  Certified by Micro1 in React, JS(ES6+), HTML5, CSS3, RESTful APIs & third-party services integration

Languages

Arabic

Native

English

Professional Working Proficiency

Education

High School Degree, *Al-amjad School - Taiz - Yemen*

Taiz, Yemen